

Here tensorflow library is used for CNN by Google.

```
import cv2
import numpy as np
import os
import zipfile

from google.colab.patches import cv2_imshow
import pandas as pd
import seaborn as sns

import matplotlib.pyplot as plt
import tensorflow as tf
from google.colab import drive

drive.mount('/content/drive')
```

Mounted at /content/drive

```
path = '/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/10_Image_Classification_usingCNN/homer_bart_1.zip'
```

```
z_object = zipfile.ZipFile(path, mode= 'r')
z_object.extractall('./')
z_object.close()
```

```
ls
```

```
drive/  homer_bart_1/  __MACOSX/  sample_data/
```

```
# Train CNN using certain images of homer and bart
```

```
#os.path.join(directory, 'bart59.bmp')
```

Start coding or [generate](#) with AI.

```
directory = '/content/homer_bart_1'
sorted(os.listdir(directory))
```

```
['.DS_Store',
 'bart1.bmp',
 'bart10.bmp',
 'bart100.bmp',
 'bart101.bmp',
 'bart102.bmp',
 'bart103.bmp',
 'bart104.bmp',
 'bart105.bmp',
 'bart106.bmp',
 'bart108.bmp',
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 'bart132.bmp',
 'bart133.bmp',
 'bart134.bmp',
 'bart135.bmp',
```

```
'bart136.bmp',
'bart137.bmp',
'bart138.bmp',
'bart139.bmp',
'bart14.bmp',
'bart141.bmp',
'bart142.bmp',
'bart143.bmp',
'bart144.bmp',
'bart145.bmp',
'bart146.bmp',
'bart147.bmp',
'bart148.bmp',
'bart149.bmp',
'bart15.bmp',
'bart150.bmp',
'bart151.bmp',
'bart153.bmp',
```

```
# joined
files = [os.path.join(directory, f) for f in sorted(os.listdir(directory))]
#files # This is actually a list.
type(files)
```

```
list
```

```
# Converting images to array
height, width = 128, 128
```

```
# 128* 128*3 # number of neurons at the input layer, if it is coloured.
# for gray scale : no.of neurons: 128*128*1
```

✓ Convert all the images to gray scale and same pixel :128*128.

```
images = []
classes = []
for im_path in files:
    try:
        image = cv2.imread(im_path)
        H,W = image.shape[:2]
    except:
        continue

    print(f"The size of the image before resizing :{image.shape}")
    image = cv2.resize(image, (width, height))
    image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    cv2_imshow(image)
    print(f"The size of the image after resizing :{image.shape}")

    image = image.ravel()
    print(f"After transforming to 1D array : {image.shape}")
    images.append(image)

    print(os.path.basename(os.path.normpath(im_path)))
    image_name = os.path.basename(os.path.normpath(im_path))
    if image_name.startswith('b'):
        class_name = 0
    else:
        class_name = 1
    classes.append(class_name)
    print(f"The above image is of class : {class_name} \n\n\n")
```


The size of the image before resizing :(497, 326, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart1.bmp

The above image is of class : 0

The size of the image before resizing :(517, 369, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart10.bmp

The above image is of class : 0

The size of the image before resizing :(435, 364, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart100.bmp

The above image is of class : 0

The size of the image before resizing :(499, 303, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart101.bmp

The above image is of class : 0

The size of the image before resizing :(434, 318, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart102.bmp

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The size of the image before resizing :(268, 315, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart103.bmp

The above image is of class : 0

The size of the image before resizing :(467, 302, 3)



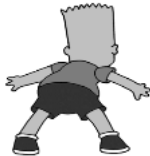
The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart104.bmp

The above image is of class : 0

The size of the image before resizing :(298, 250, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart105.bmp

The above image is of class : 0

The size of the image before resizing :(650, 340, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart106.bmp

The above image is of class : 0

The size of the image before resizing :(629, 302, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart108.bmp

The above image is of class : 0

The size of the image before resizing :(357, 400, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart109.bmp

The above image is of class : 0

The size of the image before resizing :(488, 367, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart111.bmp

The above image is of class : 0

The size of the image before resizing :(394, 330, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart110.bmp

The above image is of class : 0

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After transforming to 1D array : (16384,)

bart111.bmp

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After transforming to 1D array : (16384,)

bart112.bmp

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After transforming to 1D array : (16384,)

bart113.bmp

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The size of the image before resizing :(535, 535, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart114.bmp

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The size of the image before resizing :(714, 435, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

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After transforming to 1D array : (16384,)
bart115.bmp
The above image is of class : 0
```

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After transforming to 1D array : (16384,)
bart116.bmp
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```

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```
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```
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bart118.bmp
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```

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```
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bart119.bmp
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```

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```
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```

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bart120.bmp
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The size of the image after resizing :(128, 128)
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bart121.bmp
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The size of the image before resizing :(880, 539, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart122.bmp
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The size of the image before resizing :(726, 494, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart123.bmp
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart124.bmp
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bart127.bmp
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The size of the image after resizing :(128, 128)
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bart128.bmp
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bart13.bmp
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The size of the image before resizing :(321, 383, 3)



The size of the image after resizing :(128, 128)
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bart130.bmp
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The size of the image before resizing :(309, 510, 3)



The size of the image after resizing :(128, 128)
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bart131.bmp
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The size of the image before resizing :(293, 410, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart132.bmp
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The size of the image after resizing :(128, 128)
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bart14.bmp
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The size of the image after resizing :(128, 128)
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bart141.bmp
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The size of the image after resizing :(128, 128)
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The size of the image after resizing :(128, 128)
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bart144.bmp
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The size of the image before resizing :(701, 417, 3)



The size of the image after resizing :(128, 128)
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The size of the image before resizing :(617, 582, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image before resizing :(502, 468, 3)



The size of the image after resizing :(128, 128)
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bart147.bmp
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The size of the image before resizing :(547, 568, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart148.bmp
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The size of the image before resizing :(406, 253, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart149.bmp
The above image is of class : 0

The size of the image before resizing :(474, 412, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart15.bmp

The above image is of class : 0

The size of the image before resizing :(320, 268, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart150.bmp

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The size of the image before resizing :(447, 293, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart151.bmp

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The size of the image before resizing :(356, 280, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart153.bmp

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The size of the image before resizing :(375, 300, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart154.bmp

The above image is of class : 0

The size of the image before resizing :(555, 555, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart155.bmp

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The size of the image before resizing :(540, 380, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart156.bmp

The above image is of class : 0

The size of the image before resizing :(525, 412, 3)



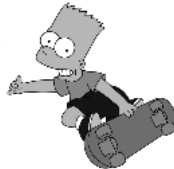
The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart157.bmp

The above image is of class : 0

The size of the image before resizing :(555, 555, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart158.bmp

The above image is of class : 0

The size of the image before resizing :(586, 411, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart159.bmp

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The size of the image before resizing :(426, 576, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart160.bmp

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The size of the image before resizing :(591, 555, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart160.bmp

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The size of the image before resizing :(588, 357, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart161.bmp

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The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart162.bmp

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The size of the image before resizing :(594, 475, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart164.bmp

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The size of the image before resizing :(620, 475, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart165.bmp

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The size of the image before resizing :(579, 401, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart166.bmp

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The size of the image before resizing :(525, 472, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)
bart167.bmp
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The size of the image before resizing :(534, 336, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart168.bmp
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The size of the image before resizing :(413, 380, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart169.bmp
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The size of the image before resizing :(477, 516, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart17.bmp
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The size of the image after resizing :(128, 128)
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bart18.bmp
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart19.bmp
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The size of the image before resizing :(515, 501, 3)



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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart2.bmp
The above image is of class : 0

The size of the image before resizing :(431, 193, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart20.bmp
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart21.bmp
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image before resizing :(533, 432, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart25.bmp
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The above image is of class : 0

The size of the image before resizing :(445, 303, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart27.bmp
The above image is of class : 0

The size of the image before resizing :(514, 428, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart28.bmp
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The size of the image before resizing :(556, 400, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart29.bmp
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The size of the image after resizing :(128, 128)
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The size of the image before resizing :(547, 426, 3)





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After transforming to 1D array : (16384,)
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The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image before resizing :(498, 371, 3)



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After transforming to 1D array : (16384,)
bart32.bmp
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The size of the image before resizing :(455, 584, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart34.bmp
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The size of the image before resizing :(597, 336, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart35.bmp
The above image is of class : 0

The size of the image before resizing :(459, 558, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart36.bmp
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The size of the image before resizing :(444, 241, 3)





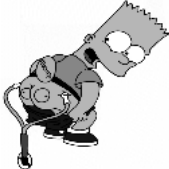
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart37.bmp
The above image is of class : 0

The size of the image before resizing :(424, 358, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart38.bmp
The above image is of class : 0

The size of the image before resizing :(601, 515, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart39.bmp
The above image is of class : 0

The size of the image before resizing :(437, 270, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart4.bmp
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The size of the image before resizing :(524, 401, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The size of the image before resizing :(517, 221, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
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The above image is of class : 0

The size of the image before resizing :(425, 280, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart43.bmp
The above image is of class : 0

The size of the image before resizing :(353, 181, 3)



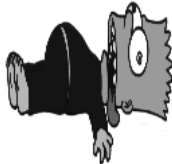
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart44.bmp
The above image is of class : 0

The size of the image before resizing :(349, 299, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart45.bmp
The above image is of class : 0

The size of the image before resizing :(220, 538, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart46.bmp
The above image is of class : 0

The size of the image before resizing :(480, 268, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart47.bmp
The above image is of class : 0

The size of the image before resizing :(344, 316, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart48.bmp
The above image is of class : 0

The size of the image before resizing :(597, 336, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart49.bmp

The above image is of class : 0

The size of the image before resizing :(531, 442, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart5.bmp

The above image is of class : 0

The size of the image before resizing :(180, 300, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart50.bmp

The above image is of class : 0

The size of the image before resizing :(340, 304, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart51.bmp

The above image is of class : 0

The size of the image before resizing :(347, 189, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart52.bmp

The above image is of class : 0

The size of the image before resizing :(350, 162, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart53.bmp

The above image is of class : 0

The size of the image before resizing :(190, 238, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart54.bmp

The above image is of class : 0

The size of the image before resizing :(350, 128, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart55.bmp

The above image is of class : 0

The size of the image before resizing :(206, 158, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart56.bmp

The above image is of class : 0

The size of the image before resizing :(357, 213, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart57.bmp

The above image is of class : 0

The size of the image before resizing :(524, 401, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart58.bmp

The above image is of class : 0

The size of the image before resizing :(430, 339, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart59.bmp

```
bart55.bmp  
The above image is of class : 0
```

The size of the image before resizing :(481, 381, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)

```
bart6.bmp  
The above image is of class : 0
```

The size of the image before resizing :(293, 251, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)

```
bart60.bmp  
The above image is of class : 0
```

The size of the image before resizing :(386, 316, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)

```
bart61.bmp  
The above image is of class : 0
```

The size of the image before resizing :(272, 309, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)

```
bart62.bmp  
The above image is of class : 0
```

The size of the image before resizing :(279, 212, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)

```
bart63.bmp  
The above image is of class : 0
```

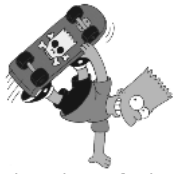
The size of the image before resizing :(281, 232, 3)



The size of the image after resizing :(128, 128)


```
After transforming to 1D array : (16384,)
bart64.bmp
The above image is of class : 0
```

The size of the image before resizing :(344, 356, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart65.bmp
The above image is of class : 0
```

The size of the image before resizing :(264, 234, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart66.bmp
The above image is of class : 0
```

The size of the image before resizing :(314, 276, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart67.bmp
The above image is of class : 0
```

The size of the image before resizing :(270, 150, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart68.bmp
The above image is of class : 0
```

The size of the image before resizing :(564, 480, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart69.bmp
The above image is of class : 0
```

The size of the image before resizing :(278, 355, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart7.bmp
The above image is of class : 0

The size of the image before resizing :(283, 228, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart70.bmp
The above image is of class : 0

The size of the image before resizing :(328, 190, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart71.bmp
The above image is of class : 0

The size of the image before resizing :(339, 275, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart72.bmp
The above image is of class : 0

The size of the image before resizing :(478, 286, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart73.bmp
The above image is of class : 0

The size of the image before resizing :(392, 229, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart74.bmp
The above image is of class : 0

The size of the image before resizing :(248, 163, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart75.bmp
The above image is of class : 0

The size of the image before resizing :(271, 163, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart76.bmp
The above image is of class : 0

The size of the image before resizing :(246, 233, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart77.bmp
The above image is of class : 0

The size of the image before resizing :(281, 154, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart79.bmp
The above image is of class : 0

The size of the image before resizing :(593, 409, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart8.bmp
The above image is of class : 0

The size of the image before resizing :(400, 345, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
bart80.bmp
The above image is of class : 0

The size of the image before resizing :(405, 214, 3)





The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart81.bmp
 The above image is of class : 0

The size of the image before resizing :(303, 207, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart82.bmp
 The above image is of class : 0

The size of the image before resizing :(349, 228, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart83.bmp
 The above image is of class : 0

The size of the image before resizing :(303, 246, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart84.bmp
 The above image is of class : 0

The size of the image before resizing :(303, 213, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart85.bmp
 The above image is of class : 0

The size of the image before resizing :(497, 333, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart86.bmp
 The above image is of class : 0

The size of the image before resizing :(384, 310, 3)





The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart87.bmp
 The above image is of class : 0

The size of the image before resizing :(360, 276, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart89.bmp
 The above image is of class : 0

The size of the image before resizing :(572, 318, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart9.bmp
 The above image is of class : 0

The size of the image before resizing :(415, 272, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart90.bmp
 The above image is of class : 0

The size of the image before resizing :(330, 400, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart91.bmp
 The above image is of class : 0

The size of the image before resizing :(430, 400, 3)



The size of the image after resizing :(128, 128)
 After transforming to 1D array : (16384,)
 bart92.bmp
 The above image is of class : 0

The size of the image before resizing :(465, 335, 3)

The size of the image before resizing :(403, 333, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart93.bmp

The above image is of class : 0

The size of the image before resizing :(190, 305, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart94.bmp

The above image is of class : 0

The size of the image before resizing :(398, 290, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart95.bmp

The above image is of class : 0

The size of the image before resizing :(475, 260, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart96.bmp

The above image is of class : 0

The size of the image before resizing :(405, 380, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart97.bmp

The above image is of class : 0

The size of the image before resizing :(476, 458, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart98.bmp

The above image is of class : 0

The size of the image before resizing :(452, 252, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

bart99.bmp

The above image is of class : 0

The size of the image before resizing :(576, 237, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer1.bmp

The above image is of class : 1

The size of the image before resizing :(380, 157, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer10.bmp

The above image is of class : 1

The size of the image before resizing :(560, 276, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer100.bmp

The above image is of class : 1

The size of the image before resizing :(620, 465, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer101.bmp

The above image is of class : 1

The size of the image before resizing :(464, 429, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer102.bmp

The above image is of class : 1

The size of the image before resizing :(335, 271, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer103.bmp

The above image is of class : 1

The size of the image before resizing :(365, 394, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer104.bmp

The above image is of class : 1

The size of the image before resizing :(349, 270, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer105.bmp

The above image is of class : 1

The size of the image before resizing :(361, 272, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer106.bmp

The above image is of class : 1

The size of the image before resizing :(359, 392, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer107.bmp

The above image is of class : 1

The size of the image before resizing :(392, 395, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer109.bmp

The above image is of class : 1

The size of the image before resizing :(352, 385, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer110.bmp

The above image is of class : 1

The size of the image before resizing :(495, 438, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer112.bmp

The above image is of class : 1

The size of the image before resizing :(488, 454, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer113.bmp

The above image is of class : 1

The size of the image before resizing :(563, 544, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer115.bmp

The above image is of class : 1

The size of the image before resizing :(418, 523, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer116.bmp

The above image is of class : 1

The size of the image before resizing :(468, 445, 3)



The size of the image after resizing :(128, 128)

```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer117.bmp
The above image is of class : 1
```

The size of the image before resizing :(522, 608, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer119.bmp
The above image is of class : 1
```

The size of the image before resizing :(491, 608, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer120.bmp
The above image is of class : 1
```

The size of the image before resizing :(725, 261, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer121.bmp
The above image is of class : 1
```

The size of the image before resizing :(628, 235, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer122.bmp
The above image is of class : 1
```

The size of the image before resizing :(551, 589, 3)



```
The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer123.bmp
The above image is of class : 1
```

The size of the image before resizing :(601, 250, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer124.bmp
The above image is of class : 1

The size of the image before resizing :(500, 242, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer13.bmp
The above image is of class : 1

The size of the image before resizing :(604, 240, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer14.bmp
The above image is of class : 1

The size of the image before resizing :(536, 514, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer15.bmp
The above image is of class : 1

The size of the image before resizing :(361, 522, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer16.bmp
The above image is of class : 1

The size of the image before resizing :(470, 533, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer17.bmp
The above image is of class : 1

The size of the image before resizing :(532, 193, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer18.bmp
The above image is of class : 1

The size of the image before resizing :(511, 388, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer2.bmp
The above image is of class : 1

The size of the image before resizing :(431, 226, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer20.bmp
The above image is of class : 1

The size of the image before resizing :(474, 252, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer21.bmp
The above image is of class : 1

The size of the image before resizing :(520, 206, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer22.bmp
The above image is of class : 1

The size of the image before resizing :(552, 283, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer23.bmp
The above image is of class : 1

The size of the image before resizing :(496, 369, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer24.bmp
The above image is of class : 1

The size of the image before resizing :(544, 520, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer26.bmp
The above image is of class : 1

The size of the image before resizing :(329, 571, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer27.bmp
The above image is of class : 1

The size of the image before resizing :(468, 392, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer28.bmp
The above image is of class : 1

The size of the image before resizing :(506, 224, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer29.bmp
The above image is of class : 1

The size of the image before resizing :(511, 350, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer3.bmp
The above image is of class : 1

The size of the image before resizing :(483, 222, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer30.bmp
The above image is of class : 1

The size of the image before resizing :(438, 335, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer31.bmp
The above image is of class : 1

The size of the image before resizing :(446, 296, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer32.bmp
The above image is of class : 1

The size of the image before resizing :(234, 257, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer33.bmp
The above image is of class : 1

The size of the image before resizing :(350, 265, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer34.bmp
The above image is of class : 1

The size of the image before resizing :(232, 191, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer35.bmp
The above image is of class : 1

The size of the image before resizing :(370, 211, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer36.bmp
The above image is of class : 1

The size of the image before resizing :(200, 305, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer37.bmp
The above image is of class : 1

The size of the image before resizing :(264, 327, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer38.bmp
The above image is of class : 1

The size of the image before resizing :(325, 273, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer4.bmp
The above image is of class : 1

The size of the image before resizing :(266, 287, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer40.bmp
The above image is of class : 1

The size of the image before resizing :(278, 284, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer41.bmp
The above image is of class : 1

The size of the image before resizing :(243, 251, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer43.bmp

The above image is of class : 1

The size of the image before resizing :(348, 267, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer44.bmp

The above image is of class : 1

The size of the image before resizing :(262, 251, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer45.bmp

The above image is of class : 1

The size of the image before resizing :(302, 244, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer46.bmp

The above image is of class : 1

The size of the image before resizing :(306, 163, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer47.bmp

The above image is of class : 1

The size of the image before resizing :(187, 225, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer48.bmp

The above image is of class : 1

The above image is of class : 1

The size of the image before resizing :(261, 225, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer49.bmp

The above image is of class : 1

The size of the image before resizing :(499, 339, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer5.bmp

The above image is of class : 1

The size of the image before resizing :(444, 353, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer50.bmp

The above image is of class : 1

The size of the image before resizing :(380, 157, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer51.bmp

The above image is of class : 1

The size of the image before resizing :(269, 380, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer52.bmp

The above image is of class : 1

The size of the image before resizing :(343, 350, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer54.bmp

The above image is of class : 1

The size of the image before resizing :(420, 255, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer55.bmp

The above image is of class : 1

The size of the image before resizing :(386, 326, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer56.bmp

The above image is of class : 1

The size of the image before resizing :(433, 329, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer57.bmp

The above image is of class : 1

The size of the image before resizing :(281, 200, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer58.bmp

The above image is of class : 1

The size of the image before resizing :(361, 250, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer59.bmp

The above image is of class : 1

The size of the image before resizing :(664, 296, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer6.bmp
The above image is of class : 1

The size of the image before resizing :(341, 250, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer60.bmp
The above image is of class : 1

The size of the image before resizing :(505, 280, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer61.bmp
The above image is of class : 1

The size of the image before resizing :(339, 406, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer62.bmp
The above image is of class : 1

The size of the image before resizing :(259, 338, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer63.bmp
The above image is of class : 1

The size of the image before resizing :(503, 300, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer65.bmp
The above image is of class : 1

The size of the image before resizing :(410, 525, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer66.bmp
The above image is of class : 1

The size of the image before resizing :(410, 310, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer68.bmp
The above image is of class : 1

The size of the image before resizing :(411, 348, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer69.bmp
The above image is of class : 1

The size of the image before resizing :(529, 325, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer7.bmp
The above image is of class : 1

The size of the image before resizing :(385, 398, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer70.bmp
The above image is of class : 1

The size of the image before resizing :(400, 426, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer71.bmp
The above image is of class : 1

The size of the image before resizing :(396, 272, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer72.bmp
The above image is of class : 1

The size of the image before resizing :(436, 334, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer73.bmp
The above image is of class : 1

The size of the image before resizing :(403, 300, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer74.bmp
The above image is of class : 1

The size of the image before resizing :(429, 503, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer75.bmp
The above image is of class : 1

The size of the image before resizing :(358, 500, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer76.bmp
The above image is of class : 1

The size of the image before resizing :(699, 360, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer77.bmp
The above image is of class : 1

The size of the image before resizing :(450, 564, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer78.bmp
The above image is of class : 1

The size of the image before resizing :(740, 450, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer79.bmp
The above image is of class : 1

The size of the image before resizing :(554, 472, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer8.bmp
The above image is of class : 1

The size of the image before resizing :(468, 430, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer80.bmp
The above image is of class : 1

The size of the image before resizing :(528, 268, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer81.bmp
The above image is of class : 1

The size of the image before resizing :(585, 510, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer82.bmp
The above image is of class : 1

The size of the image before resizing :(540, 468, 3)





The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer83.bmp
The above image is of class : 1

The size of the image before resizing :(516, 590, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer84.bmp
The above image is of class : 1

The size of the image before resizing :(413, 473, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer85.bmp
The above image is of class : 1

The size of the image before resizing :(337, 267, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer86.bmp
The above image is of class : 1

The size of the image before resizing :(305, 253, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer87.bmp
The above image is of class : 1

The size of the image before resizing :(607, 495, 3)



The size of the image after resizing :(128, 128)
After transforming to 1D array : (16384,)
homer9.bmp
The above image is of class : 1

The size of the image before resizing :(600, 490, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer90.bmp

The above image is of class : 1

The size of the image before resizing :(469, 354, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer91.bmp

The above image is of class : 1

The size of the image before resizing :(437, 194, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer92.bmp

The above image is of class : 1

The size of the image before resizing :(390, 533, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer93.bmp

The above image is of class : 1

The size of the image before resizing :(387, 328, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer94.bmp

The above image is of class : 1

The size of the image before resizing :(448, 366, 3)



The size of the image after resizing :(128, 128)

After transforming to 1D array : (16384,)

homer95.bmp

The above image is of class : 1


```
#cv2.imread(files[2])
```

```
The size of the image before resizing :(373, 413, 3)
```

```
#cv2.imread(files[4])
```

Double-click for enter) to edit

images

```
After transforming to 1D array: (16384,) dtype=uint8),
home99.bmp, 255, 254, ..., 254, 254, 254], dtype=uint8),
The above image is of class 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 128, 123, 124], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
The size of the image before resizing: (255, 255, 3)
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([254, 254, 254, ..., 254, 254, 254], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
The size of the image after resizing: (255, 255, 3) dtype=uint8),
After transforming to 1D array: (16384,) dtype=uint8),
home97.bmp, 255, 255, ..., 255, 255, 255], dtype=uint8),
The above image is of class 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 252, 255, 255], dtype=uint8),
The size of the image before resizing: (255, 255, 3) dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
The size of the image after resizing: (255, 255, 3) dtype=uint8),
After transforming to 1D array: (16384,) dtype=uint8),
home98.bmp, 255, 255, ..., 255, 255, 255], dtype=uint8),
The above image is of class 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 252, 253], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
The size of the image before resizing: (255, 255, 3) dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 254], dtype=uint8),
The size of the image after resizing: (255, 255, 3) dtype=uint8),
After transforming to 1D array: (16384,) dtype=uint8),
home99.bmp, 255, 255, ..., 255, 255, 255], dtype=uint8),
The above image is of class 253, 254, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 252, 255, 255], dtype=uint8),
array([254, 254, 254, ..., 254, 254, 254], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
array([255, 255, 255, ..., 255, 255, 255], dtype=uint8),
```

classes

```
[0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
```


Prevents activation functions from saturating.

- Min-Max Scaling (Used when there is no outliers.)
- z normalization/ z-scoring/standardization.

[illegible]

```
# since we dont have any outliers, Iam going for minmax scaling.
from sklearn.preprocessing import MinMaxScaler
sc = MinMaxScaler()
X = sc.fit_transform(x)
```

- ✓ Train test split.

- At input layer
- At hidden layers : $16384/2 + 1$

51/56

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 8193)	134,242,305
dense_1 (Dense)	(None, 8193)	67,133,442
dense_2 (Dense)	(None, 1)	8,194

Total params: 201,383,941 (768.22 MB)

Trainable params: 201,383,941 (768.22 MB)

Non-trainable params: 0 (0.00 B)

Next step, Compile the optimizer.

Need to specify the optimizer, That is the algorithm to optimize the weights.

```
network1.compile(optimizer= 'Adam', loss='binary_crossentropy', metrics= ['accuracy'])
```

```
history = network1.fit(X_train, y_train, epochs= 75)
```

```
Epoch 1/75
7/7 ----- 31s 3s/step - accuracy: 0.4641 - loss: 161.5041
Epoch 2/75
7/7 ----- 38s 4s/step - accuracy: 0.4258 - loss: 7.8864
Epoch 3/75
7/7 ----- 42s 4s/step - accuracy: 0.5333 - loss: 1.6532
Epoch 4/75
7/7 ----- 38s 3s/step - accuracy: 0.5706 - loss: 1.8220
Epoch 5/75
7/7 ----- 42s 3s/step - accuracy: 0.5466 - loss: 2.0951
Epoch 6/75
7/7 ----- 40s 3s/step - accuracy: 0.5904 - loss: 1.2690
Epoch 7/75
7/7 ----- 41s 3s/step - accuracy: 0.5542 - loss: 0.7289
Epoch 8/75
7/7 ----- 41s 3s/step - accuracy: 0.5821 - loss: 0.6823
Epoch 9/75
7/7 ----- 41s 3s/step - accuracy: 0.5728 - loss: 0.6852
Epoch 10/75
7/7 ----- 48s 4s/step - accuracy: 0.6006 - loss: 0.6493
Epoch 11/75
7/7 ----- 33s 5s/step - accuracy: 0.7586 - loss: 0.6042
Epoch 12/75
7/7 ----- 33s 3s/step - accuracy: 0.5852 - loss: 0.6806
Epoch 13/75
7/7 ----- 23s 3s/step - accuracy: 0.6291 - loss: 0.6097
Epoch 14/75
7/7 ----- 25s 3s/step - accuracy: 0.6141 - loss: 0.6507
Epoch 15/75
7/7 ----- 24s 3s/step - accuracy: 0.6942 - loss: 0.6040
Epoch 16/75
7/7 ----- 24s 3s/step - accuracy: 0.6672 - loss: 0.5665
Epoch 17/75
7/7 ----- 24s 3s/step - accuracy: 0.6878 - loss: 0.5834
Epoch 18/75
7/7 ----- 40s 3s/step - accuracy: 0.6127 - loss: 0.6904
Epoch 19/75
7/7 ----- 41s 3s/step - accuracy: 0.6881 - loss: 0.6194
Epoch 20/75
7/7 ----- 41s 3s/step - accuracy: 0.7676 - loss: 0.4979
Epoch 21/75
7/7 ----- 41s 3s/step - accuracy: 0.6893 - loss: 0.6929
Epoch 22/75
7/7 ----- 41s 3s/step - accuracy: 0.6542 - loss: 0.6082
Epoch 23/75
7/7 ----- 25s 3s/step - accuracy: 0.7821 - loss: 0.4993
Epoch 24/75
7/7 ----- 23s 3s/step - accuracy: 0.7452 - loss: 0.5007
Epoch 25/75
7/7 ----- 25s 3s/step - accuracy: 0.7226 - loss: 0.5456
Epoch 26/75
7/7 ----- 41s 4s/step - accuracy: 0.7856 - loss: 0.4672
Epoch 27/75
7/7 ----- 23s 3s/step - accuracy: 0.7879 - loss: 0.4230
Epoch 28/75
7/7 ----- 41s 3s/step - accuracy: 0.8627 - loss: 0.3329
Epoch 29/75
7/7 ----- 41s 3s/step - accuracy: 0.7740 - loss: 0.5066
```

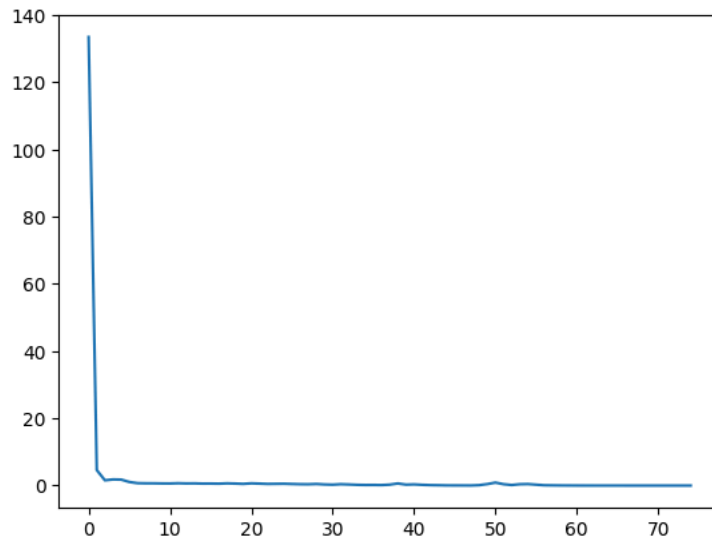
✓ Evaluating the neural network

```
history.history.keys()
```

```
dict_keys(['accuracy', 'loss'])
```

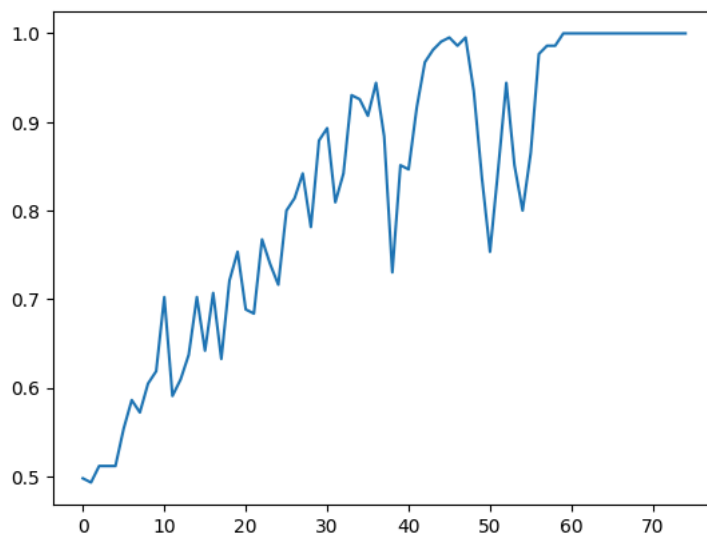
```
plt.plot(history.history['loss'])
```

```
[<matplotlib.lines.Line2D at 0x7fc50df39d00>]
```



```
plt.plot(history.history['accuracy'])
```

```
[<matplotlib.lines.Line2D at 0x7fc506e63f50>]
```



```
predictions = network1.predict(X_test)
```

2/2 ————— 2s 523ms/step

```
predictions
```

```
array([[9.9309301e-01],  
       [5.1013953e-03],  
       [9.8851871e-01],  
       [8.6638965e-03],  
       [8.3646573e-02],  
       [9.9969953e-01],  
       [4.9685203e-03],  
       [9.9998784e-01],  
       [9.5278782e-01],  
       [5.6255911e-03],  
       [1.2323172e-04],  
       [9.9269640e-01],  
       [2.3195483e-03],  
       [3.4480374e-02],  
       [7.9061192e-01],  
       [1.5726684e-02],  
       [6.6048675e-04],  
       [3.1383347e-03],  
       [1.9567488e-02],  
       [4.8100719e-01],  
       [9.2918676e-01],  
       [8.3906998e-05],
```

```
array([[ True],
       [False],
       [ True],
       [False],
       [False],
       [ True],
       [ True],
       [False],
       [False],
       [ True],
       [False],
       [False],
       [ True],
       [False],
       [False],
       [False],
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       [False],
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       [False],
       [ True],
       [ True],
       [ True],
       [ True],
       [ True],
       [ True],
       [False],
       [ True],
       [ True],
       [False]])
```

```
[False],
[False]])
```

```
y_test
```

```
array([[1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0,
        0, 1, 1, 1, 0, 1, 1, 1, 0, 0]])
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
accuracy_score(y_test, predictions)
```

```
0.6851851851851852
```

```
confusion_matrix(y_test, predictions)
```

```
array([[25,  5],
       [12, 12]])
```

```
print(classification_report(y_test, predictions))
```

	precision	recall	f1-score	support
0	0.68	0.83	0.75	30
1	0.71	0.50	0.59	24
accuracy			0.69	54
macro avg	0.69	0.67	0.67	54
weighted avg	0.69	0.69	0.67	54

```
# Saving and Loading the trained network
```

Save

```
model_json = network1.to_json()
with open('/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/10_Image_Classification_usingCNN/network1.json'
        as json_file.write(model_json)
```

```
from keras.models import save_model
```

```
save_model(network1, '/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/10_Image_Classification_usingCNN/weig
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. Th
```

Start coding or [generate](#) with AI.

Load the model

```
with open('/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/10_Image_Classification_usingCNN/network1.json'
        as json_file.read())
```

```
jason_saved_model
```

```
{'module': 'keras', 'class_name': 'Sequential', 'config': {'name': 'sequential', 'trainable': true, 'dtype':
{'module': 'keras', 'class_name': 'DTypePolicy', 'config': {'name': 'float32'}, 'registered_name': null, 'layer
s': [{'module': 'keras.layers', 'class_name': 'InputLayer', 'config': {'batch_shape': [null, 16384], 'dtype': 'f
loat32', 'sparse': false, 'ragged': false, 'name': 'input_layer'}, 'registered_name': null}, {'module': 'keras.l
ayers', 'class_name': 'Dense', 'config': {'name': 'dense', 'trainable': true, 'dtype': {'module': 'keras', 'clas
s_name': 'DTypePolicy', 'config': {'name': 'float32'}, 'registered_name': null, 'units': 8193, 'activation': 'r
elu', 'use_bias': true, 'kernel_initializer': {'module': 'keras.initializers', 'class_name': 'GlorotUniform', 'c
```

```
network_loaded = tf.keras.models.model_from_json(jason_saved_model)
network_loaded.load_weights('/content/drive/MyDrive/MCV_Continuatin_inGoogle_Colab/10_Image_Classification_using
# compile
network_loaded.compile(optimizer= 'Adam', loss='binary_crossentropy', metrics= ['accuracy'])
```

```
network_loaded.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 8193)	134,242,305
dense_1 (Dense)	(None, 8193)	67,133,442
dense_2 (Dense)	(None, 1)	8,194

Total params: 201,383,941 (768.22 MB)

Trainable params: 201,383,941 (768.22 MB)

Non-trainable params: 0 (0.00 B)

✓ Testing for new image.

```
newImage = X_test[10]  
X_test[10].shape
```

```
(16384,)
```

```
testImage = sc.inverse_transform(newImage.reshape(1,-1))  
testImage
```

```
array([[255., 255., 255., ..., 255., 255., 255.]])
```